Retrieving Relations, Detecting Fallacies: A RAG Approach to Political Debate Analysis

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Abstract

Fallacies are arguments that employ invalid reasoning, making their automatic detection critical in sensitive contexts such as high-stakes political debates, where public opinion is shaped. Spotting a fallacious argument requires contextual knowledge beyond its pure surface text. This entails world knowledge pertaining to the subject matter under discussion, as well as knowledge of the relationships that exist between arguments within the argumentative discourse. Prior work on fallacy analysis has shown that argumentative discourse structure can beneficially improve classification performance. However, such structure is typically encoded only as static classifier features, limiting its flexibility. Building on this intuition while addressing this limitation, we introduce a guided retrieval-augmented methodology for fallacy detection and classification that leverages argumentative relations of support and attack to dynamically steer the extraction of relevant documents. We evaluate our approach on the *ElecDeb60to20* benchmark across 42 retrieval configurations and 14 models, performing retrieval over a 15GB knowledge base of collected political-related documents. Our approach improves macro-F1 up to 0.864 for fallacy detection and up to 0.725 for classification over non-retrieval baselines. These results show that incorporating external knowledge significantly enhances fallacy detection and classification when retrieval is argumentatively guided.

1 Introduction

Fallacies are argumentative moves that appear valid but violate the principles of correct reasoning (Eemeren, 2001). Though these misleading arguments sound convincing on the surface, their underlying aim is to manipulate public opinion. In this light, recognizing fallacies becomes crucial, to enable citizens to engage critically with political information, spot misleading rhetorical strategies, and evaluate candidate claims (Abbas et al., 2024). A prominent example is *appeal to fear*, where speakers exaggerate hypothetical dangers to trigger an emotional response rather than presenting sound evidence (Shand, 1995).

Given the potential negative impact of fallacious rhetoric on public discourse, identifying and classifying these arguments remains a key challenge in Argument Mining (AM)(Stab and Gurevych, 2017; Sahai et al., 2021; Ruiz-Dolz and Lawrence, 2023). Recent studies employed argumentative relations as encoded features to improve fallacy detection and classification (Goffredo et al., 2023) results. However, structural information alone is often insufficient to identify fallacious arguments, as it requires to be integrated with world knowledge, going beyond the surface text of a argument. To tackle this issue, we introduce a novel method to address the tasks of fallacy detection and classification through *argumentative relation guided retrieval*. Our approach leverages the underlying argumentative structure, specifically the *support* and *attack* relations between argument components, to dynamically steer external knowledge retrieval. The originality of our approach lies in the fact that, rather than using argumentative structures merely as static input features, we leverage argumentative relations to retrieve highly targeted contextual evidence. Given that we target the political domain, we evaluate our framework on the ElecDeb60to20 dataset (Goffredo et al., 2023), a comprehensive benchmark spanning televised U.S. presidential debates from 1960 to 2020, annotated with argument components, relations, and fallacies. The main contributions of this paper are: **1) Relation Guided Retrieval for Fallacy Detection and Classification:** we propose a novel method that leverages argumentative relations to steer external knowledge retrieval dynamically, shifting from static structural encodings to context-aware retrieval. We test our approach on the detection of whether an argument

is a fallacy or not, and its classification into one of the six most common fallacy types. **2) A 15 GB Temporally Grounded Domain Knowledge Base**, tailored to our specific application to U.S. political debates, across six key domains (legislation, entities, geography, history, fallacy taxonomy, and labeled examples). Every record is timestamped, enabling temporally constrained retrieval to improve the focus of historical context. **3) Extensive Evaluation** of our method across 42 retrieval configurations and 14 models, showing significant gains over state-of-the-art baselines.

2 Related Work

Fallacy detection and classification. In the last years, the NLP community has shown an increasing interest in the task of fallacy detection and classification, and related challenges, such as misinformation and propaganda.

Concerning classification, Habernal et al. (2017) released *Argotario*, an open-source software, that serves as a gaming platform for educational purposes and as a crowd-sourcing data-acquisition platform to annotate fallacy types in everyday argumentation. Following this study, they released an annotated dataset of fallacious arguments in German and English, and use Support Vector Machine and BiLSTM models to classify fallacy types (Habernal et al., 2018). A parallel line of research targets propaganda and persuasion techniques, which partly overlap with logical fallacies (Martino et al., 2019). In this work, the authors define an annotation scheme of 18 propaganda techniques and annotate 451 news articles with such labels, and propose a multi-granularity network architecture on top of BERT contextualized embeddings to identify propagandist arguments on different levels of granularity. Jin et al. (2022) use an architecture based on a simple classifier that includes fallacies' structural information, achieving 0.58 F1 Score on the classification of 13 fallacy classes. Alhindi et al. (2022) propose a multitask prompting approach based on the T5 model to identify 28 fallacy types. Goffredo et al. (2022) propose a transformer-based neural architecture for detecting and classifying six types of fallacies in U.S. Presidential Election Debates from 1960 to 2016. In subsequent work, Goffredo et al. (2023) extend this approach by integrating textual, argumentative, and engineered features into a transformer-based architecture, while expanding the original dataset into *ElecDeb60to20* and consolidating the six-class fallacy taxonomy.

Concerning detection, Vorakitphan et al. (2021) propose a pipeline for propaganda detection and technique classification, combining semantic and argumentative features, using a BERT-based binary classifier, and a RoBERTa-based model, achieving 0.72 and 0.64 F1 Score, respectively, for 14 class sentence-span classification. Similarly, Sahai et al. (2021) address fallacy detection through comment- and token-level classification across eight fallacy types, using Reddit data and contextual information from parent comments. Their fine-tuned BERT achieves 0.53 Macro-F1 on token classification.

With the advancement of Large Language Models (LLMs), more recent work has started to include their knowledge for improving fallacy classification. Sourati et al. (2023) propose a three-stage framework for fallacy detection, coarse- and fine-grained classification, combining language models with background knowledge and explainable reasoning, highlighting the difficulty of fallacy identification and the need for specialized reasoning across fallacy types. Jeong et al. (2025) use an LLM prompting strategy that enriches input text by incorporating arguments with implicit context and rank contextual queries by confidence, achieving F1 of 0.57 in zero-shot and 0.45 in fine-tuned settings across 29 fallacy types. Teo et al. (2025) evaluate a set of LLMs for logical fallacy detection, finding that they perform well on simple fallacies but struggle with more complex, interpretive categories, highlighting the need for improved prompting, fine-tuning, and contextual data. Finally, Papadopulos et al. (2026) combine abstract logical structures with contextual linguistic cues for LLM-based fallacy classification, using data-driven pattern extraction from examples and explanations, improving zero- and one-shot performance.

In all these cases, however, the added knowledge is hand-crafted or model-generated, never retrieved from an external corpus.

Retrieval Augmented Generation (RAG). Lewis et al. (2020) introduce RAG as a framework for grounding language models in external knowledge retrieved at inference time. While initially developed for knowledge-intensive generation, RAG has since been extended to discriminative tasks, including text classification, where retrieved evidence can provide information beyond the model's parametric knowledge (Gao et al., 2023). Retrieval has also been applied to tasks closely

related to fallacy detection, particularly fact checking and argumentation. In fact-checking, benchmarks such as FEVER (Thorne et al., 2018) and PubHealth (Kotonya and Toni, 2020) use retrieved evidence to support claim verification. More recently, Dhole et al. (2025) introduced Retrieval-Augmented Argumentation, showing how external information can be retrieved to support argument quality and groundedness. However, fallacy detection and classification have largely relied on fine-tuned or few-shot models without explicitly incorporating external knowledge (Larumbe and Vendrell, 2025).

In this work, we investigate whether external knowledge can improve fallacy detection and classification in the specific application of political arguments. We adapt RAG to retrieve political knowledge relevant to the argumentative structure of each instance, using information from the relations between argument components to formulate targeted retrieval queries. This allows the model to condition its predictions on task-relevant external evidence rather than relying solely on its parametric knowledge.

3 Methodology

We consider fallacy detection and fallacy classification as separate tasks to assess how external knowledge contributes to each. In fallacy detection, given a sentence, the goal is to determine whether it contains a fallacious argument. In fallacy classification, given a sentence known to be fallacious, the goal is to assign it to exactly one of six fallacy types: Ad Hominem, Appeal to Authority, Appeal to Emotion, Slippery Slope, False Cause, or Slogan. To address these tasks, we propose a four-stage pipeline comprising: *(1)* query generation, *(2)* retrieval, *(3)* input generation, and *(4)* model prediction (Figure 1).

The **query generation** stage progressively incorporates information from the argumentative structure, resulting in three query variants:

(i) sentence-only query: the baseline query consists of the target sentence whose fallaciousness or fallacy type is to be predicted. When no relation is annotated, only this query strategy applies;

(ii) sentence+relations query: since identifying a fallacy often depends on its argumentative context, we exploit the relation layer. For each component, we append its related components to the target sentence, thereby providing the retriever with a broader view of the local argument;

(iii) sentence+relations+labels query: we further incorporate the labels of the argumentative relations (support or attack). Rather than representing these labels as bare tokens, we verbalise each relation as an explicit clause of the form `<related> supports/attacks <target>`, thereby encoding the direction of the argumentative relation directly.

During the **retrieval** stage, the retriever uses the generated query to retrieve relevant external evidence from the knowledge base (KB). The retrieved evidence is then combined with the local debate context to form a **single model input**, which is provided to the model for **prediction**. The proposed pipeline is not tied to a specific domain and can be extended to other settings where the relevant argumentative relations are available.

4 Resources for Fallacy Analysis of Political Debates

In the following, we describe the dataset we used in the experimental session and the KB we have created for evidence retrieval.

4.1 Dataset

To evaluate the proposed method, we choose the **ElecDeb60to20** dataset [1], to the best of our knowledge, the only available resource for political debates annotated with argumentative components (15,093 claims and 13,623 premises), argumentative component relations (20,805 support and 3,701 attack relations), and argumentative fallacies. The dataset contains the annotations for the six most common types of fallacies in political debate: **Ad Hominem** (an excessive attack on the arguer rather than their position, 200 examples), **Appeal to Emotion** (loading the argument with emotional language to exploit the audience, 924 examples), **Appeal to Authority** (citing an authority, a non-expert, or majority acceptance in place of relevant evidence, 290 examples), **Slippery Slope** (claiming an unlikely, exaggerated outcome will follow an act, 72 examples), **False Cause** (mistaking correlation for causation, 105 examples), and **Slogans** (a brief, striking phrase meant to excite the audience, 42 examples).

4.2 KB

Data collection. To ground retrieval, we first assembled a collection of documents relevant to the

[1] https://github.com/MARIANNE-INRIA/ElecDeb60to20

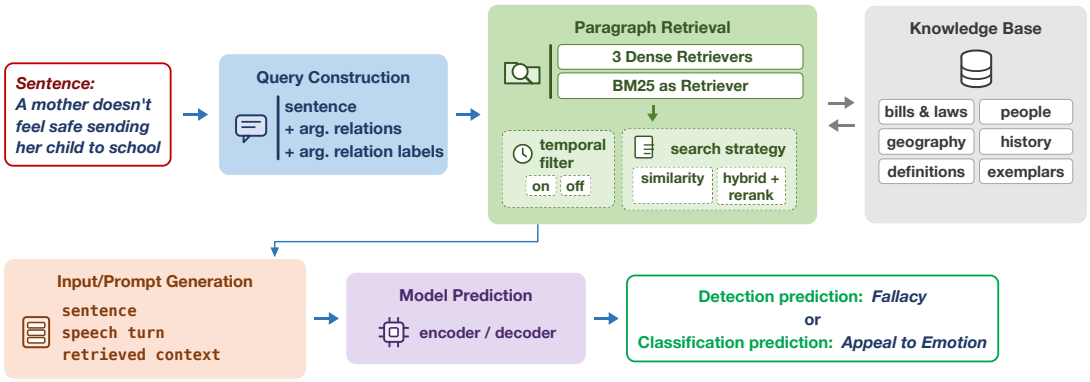


Figure 1: Proposed pipeline for fallacy analysis using external knowledge.

political scenario to create the KB.

We analysed the debate transcripts of the above mentioned dataset along two axes. First, we run the spaCy `en_core_web_trf` named-entity recogniser over every debate and group the recognised mentions into people, organisations, events and places; after normalisation and de-duplication this yields 1,081 people, 974 organisations, 80 events and 595 places. Second, we perform **topic extraction**: we segment each debate into non-overlapping 1,024-token windows, prompt Qwen2.5-7B-Instruct (Yang et al., 2025) to return the main topics of each window in a fixed `<TOPICS: t1, t2, ...>` format, parse the generations with a regular expression, and lower-case and merge them across windows. This gives 2,057 unique topics and 3,115 distinct (topic, debate-year) pairs. This LLM was chosen due to proven good capabilities in topic extraction (Li et al., 2025).

Guided by this analysis we assemble six collections of documents, each targeting a distinct kind of grounding: [2] **Legislation** comes from the U.S. Government Information website [3], whose search API we query by (topic, year), expanding each topic over a five-year window $[y-4, y]$ and keeping the 15 highest-scoring granules per pair, so as to retrieve the bills and public laws a debate of year y could plausibly be about. We complement these with the 2025 United States Government Manual and the Universal Declaration of Human Rights.

The **people** collection grounds the debates in the actors they name, combining official biographies from the Biographical Directory of the United States Congress [4] with the Wikipedia [5] pages of the individuals mentioned in the debates, expanded with the first-level pages linked from each article.

The **geography** collection draws on official U.S. Census Bureau state and territory profiles [6], including demographic, economic and social statistics, together with the Wikipedia pages of the places named in the debates and their linked pages.

For temporal grounding, the **history** collection covers U.S. history from 1900 to 2026 with one record per year, extracted from the *Events* section of the corresponding *"{year} in the United States"* Wikipedia page and complemented by the era-overview pages of *History of the United States* and their first-level links. Every record is annotated with its year, so each chunk keeps its temporal context after splitting.

The **definitions** collection supplies the theory of fallacy itself: scholarly articles published from 2015 onwards on fallacies, argumentation, rhetoric and political discourse, retrieved from the OpenAlex academic index [7], restricted to `type:article` and filtered by lexical relevance over title and abstract, together with the Wikipedia *Fallacy* page and its relevant linked articles.

Finally, the **exemplars** collection holds labelled fallacy instances drawn from LOGIC (Jin et al., 2022) and MAFALDA (Helwe et al., 2024), mapped onto our six-type taxonomy. Because these records carry gold fallacy labels, we verify that no exemplar text overlaps an ElecDeb60to20 instance and remove any that does, so that retrieval cannot leak labels into the evaluation.

Table 1 reports the resulting statistics. The collection is dominated in volume by **bills**, which account for the vast majority of both its size and its paragraphs, a direct consequence of the length and

[2] All material was crawled between May and June 2026.
[3] https://www.govinfo.gov
[4] https://bioguide.congress.gov
[5] https://en.wikipedia.org
[6] https://data.census.gov
[7] https://api.openalex.org

Collection	Size	# Documents	# Paragraphs
Bills	14 GB	178,383	6,668,178
Geography	516 MB	125,891	125,891
People	297 MB	72,950	72,950
History	89 MB	16,121	16,121
Exemplars	5 MB	1,273	1,284
Definitions	1.3 MB	65	65
Total	**15 GB**	**394,683**	**6,884,489**

Table 1: Statistics of the collected KB documents.

density of legislative text. **Geography** and **people** contribute the next largest document counts, each with a single paragraph per record. **History**, **exemplars** and **definitions** are comparatively tiny, but play a targeted role, supplying temporal grounding and fallacy-specific knowledge respectively.

Indexing. We index the KB with LangChain [8]. Every document is normalised into a JSON record with a unique identifier (`id`), its `text`, and the metadata used later to constrain retrieval: `source`, `URL`, `year` and `category` (the collection).

Long documents are split into chunks of at most 1,000 tokens with a 10% overlap. We chunk by tokens rather than by characters and clamp the chunk size to the input window of the embedding model in use, so that no passage is silently truncated at encoding time. Chunks are embedded and indexed in Chroma [9], one store per embedding model, with the document set held fixed so that retrievers can be compared in isolation.

Retrievers. We compare three dense encoders against a lexical baseline. **Sentence BERT (SBERT)** (Reimers and Gurevych, 2019) encodes queries and passages into embeddings optimised for semantic similarity. **BAAI General Embedding (BGE)** (Xiao et al., 2024) is trained on a large multi-task retrieval corpus and provides a stronger general-purpose semantic space. As a third dense retriever, we decided to pre-train a sentence transformer tailored to our political domain: **Sentence-RooseBERT**, trained with the corpus and hyperparameters defined by Dore et al. (2025). **BM25** (Robertson and Zaragoza, 2009) ranks passages by exact term overlap.

Search strategies. At inference, we retrieve the top k=5 chunks per query (as preliminary experiments showed this is the optimal number). In the **similarity** setting these come directly from the dense index by cosine similarity. In the **hybrid** setting we fuse the dense and BM25 rankings with Reciprocal Rank Fusion (k_{RRF}=60) over a pool of 30 candidates per leg, then re-rank the fused pool with a `ms-marco-MiniLM-L-6-v2` cross-encoder before keeping the top 5. Each of the three dense retrievers is evaluated with the two search strategies while BM25 is evaluated with its lexical search strategy. Each retriever is then evaluated with the three query formulations.

Here we introduce a *temporal filtering* constraint: we test whether restricting the KB to the years preceding each debate helps retrieve more focused, contextually appropriate knowledge.

5 Experimental Setting

5.1 Tasks

For the task of **fallacy detection**, we sentence-split the speech turns with spaCy and propagate the span-level annotations down to the sentence level, labelling a sentence *Fallacy* if it overlaps at least with one annotated fallacy span and *NoFallacy* otherwise. The resulting distribution is strongly skewed towards the negative class, so we downsample *NoFallacy* to a 1:1 ratio and re-split the balanced pool 80/10/10, stratified by label, yielding 2,612 training, 327 validation and 327 test sentences.

In **fallacy classification**, the task is restricted to the fallacious portion of the corpus (1,306 training, 164 validation and 163 test sentences) and remains sharply imbalanced: *Appeal to Emotion* alone accounts for more than half of the training instances (739), while *Slogans* accounts for 34. Classification is therefore the harder of the two tasks.

5.2 Models

On both tasks, we evaluate **encoder-only** models fine-tuned as sentence classifiers, and **decoder-only** LLms that output the label as text. For encoder-only models, the input is formatted as: `sentence [SEP] speech turn context`. In particular, we evaluated RoBERTa (Liu et al., 2019), DeBERTa-V3 (He et al., 2023), ModernBERT (Warner et al., 2025), NeoBERT (Breton et al., 2025) and Longformer (Beltagy et al., 2020), alongside two specialised encoders: LegalBERT (Zheng et al., 2021), chosen because a large share of our KB is legislative text, and RooseBERT (Dore et al., 2025), pre-trained on political

[8] https://www.langchain.com
[9] https://www.trychroma.com

Model	Detection			Classification		
	Baseline	RAG	Δ	Baseline	RAG	Δ
RoBERTa-base	74.7 ± 0.8	80.9 ± 1.5	$+6.2$	34.5 ± 25.0	68.0 ± 2.3	$+33.5$
DeBERTa-v3-base	74.2 ± 6.1	77.5 ± 2.2	$+3.4$	43.5 ± 13.0	45.5 ± 19.7	$+2.0$
ModernBERT-base	76.4 ± 1.7	$\mathbf{86.4 \pm 1.2}$	$+10.0$	57.7 ± 2.1	$\mathbf{72.5 \pm 4.0}$	$+14.8$
NeoBERT	57.2 ± 21.9	50.8 ± 24.0	-6.4	24.9 ± 6.8	10.8 ± 9.8	-14.1
LegalBERT	71.7 ± 1.5	77.4 ± 1.7	$+5.8$	53.9 ± 23.7	60.8 ± 3.7	$+6.9$
Longformer-base-4096	74.5 ± 2.0	78.7 ± 1.9	$+4.2$	63.3 ± 2.0	58.2 ± 4.8	-5.1
RooseBERT-scr-cased	$\mathbf{77.2 \pm 1.2}$	78.8 ± 1.1	$+1.5$	63.0 ± 2.4	69.1 ± 2.2	$+6.1$
RooseBERT-scr-uncased	69.5 ± 0.9	74.3 ± 1.8	$+4.7$	40.1 ± 4.6	27.0 ± 2.7	-13.0
RooseBERT-cont-cased	74.7 ± 2.3	77.9 ± 1.1	$+3.1$	$\mathbf{65.3 \pm 4.3}$	68.8 ± 1.5	$+3.5$
RooseBERT-cont-uncased	74.2 ± 2.5	77.0 ± 2.7	$+2.8$	44.2 ± 12.4	54.6 ± 24.4	$+10.4$
Llama-3.1-8B-Instruct	71.7 ± 0.8	52.4 ± 4.8	-19.3	53.3 ± 1.9	20.4 ± 3.0	-33.0
Mistral-7B-Instruct-v0.3	57.6 ± 0.0	43.5 ± 7.2	-14.1	34.3 ± 12.9	20.4 ± 0.0	-13.9
Qwen3.5-9B-Base	37.8 ± 9.4	33.4 ± 0.0	-4.4	3.6 ± 0.0	1.6 ± 1.0	-2.0
gpt-oss-20b	66.7 ± 1.4	54.1 ± 3.9	-12.6	31.7 ± 6.4	21.9 ± 3.3	-9.8

Table 2: Macro-F1 (mean $\pm$ std, 5 seeds) results. RAG uses a Sentence-RooseBERT retriever with hybrid search (temporal context for detection, none for classification). Δ = RAG $-$ Baseline. Best model per column in **bold**.

and parliamentary debate transcripts. RooseBERT is evaluated in all four released variants (`scr`/`cont` $\times$ cased/uncased), giving ten encoders in total. Encoders are fine-tuned over a grid of batch size $\{8, 16\}$, learning rate $\{2, 3, 5\} \times e^{-5}$ and $\{3, 5\}$ epochs, with a warmup ratio of 0.06.

For decoder-only models, a structured chat prompt includes: a system message with the task and taxonomy (for classification), the target sentence, the debate turn context, the retrieved context, and an instruction to output a single label. We evaluated `Llama-3.1-8B-Instruct` (Grattafiori et al., 2024), `Mistral-7B-Instruct-v0.3` (Jiang et al., 2023), `Qwen3.5-9B-Base` (Yang et al., 2025) and `gpt-oss-20b` (OpenAI, 2025). Each is run in three settings: `zero-shot`, `few-shot` and `fine-tune`. In `zero-shot` the model sees only the task instruction and the input. In `few-shot` we prepend a class-balanced set of labelled examples drawn from the training split, one per class. In the last setting, we perform parameter-efficient supervised fine-tuning with LoRA (Hu et al., 2022) (r=16, α=32, dropout 0.05) over learning rate $\{1, 3, 5\} \times e^{-4}$ and $\{1, 2, 3\}$ epochs at an effective batch size of 8.

5.3 Implementation

Combining retrievers, search strategies, query formulations and the temporal constraints yields 42 configurations. Adding the no-retrieval baseline gives **43 configurations per task**. Evaluating 43 configurations $\times$ 14 models $\times$ a hyper-parameter grid $\times$ five seeds is not computationally feasible, so we proceed as follows: *(i)* Every model is trained on the no-retrieval baseline across the full hyper-parameter grid. The best configuration per model is then re-run with five seeds. *(ii)* The three strongest baseline models per task are then run over all remaining 42 retrieval configurations, identifying the best retrieval configuration for that task. *(iii)* Every model is finally re-run on the best setup, sweeping over hyper-parameters and then five seeds.

All experiments use Hugging Face with Python 3.10. Encoders were trained on one H100 GPU and the LLMs on two A100 GPUs. To pre-train Sentence-RooseBERT we used eight A100 GPUs and the same hyper-parameters defined by Dore et al. (2025). Code is available at `https://anonymous.4open.science/r/FallacyRAG-0E25`, datasets as supp. material.

5.4 Metrics

Both tasks are evaluated with Accuracy and macro-averaged Precision, Recall and F1. We take **macro F1** as the primary metric, since it weights every class equally. All scores are averaged over five runs (seeds $\{3, 14, 42, 55, 89\}$), and we report mean and standard deviation. Differences between the best baseline and retriever model are tested with a two-sided paired t-test over the five paired per-seed macro-F1 values, reported with Cohen's d.

6 Results

Table 2 reports the results of the two tasks.

Fallacy Detection. Without retrieval, the strongest model is `RooseBERT-scr-cased` (0.772 macro F1), followed by `ModernBERT` (0.764) and `RooseBERT-cont-cased` (0.747). The best

of the 42 configurations combines temporal filtering, `Sentence-RooseBERT` as retriever, hybrid search with cross-encoder reranking, and *sentence+relations+labels* as the query. Under it, `ModernBERT` reaches $\mathbf{0.864}$ macro F1, $+0.100$ over its own baseline and $+0.092$ over the best baseline model. A two-sided paired t-test over the five seeds confirms the difference is significant ($t = 19.17$, $p = 4 \times 10^{-5}$, Cohen's $d = 8.57$).

Retrieval improves every encoder we test, from $+0.016$ (`RooseBERT-scr-cased`) to $+0.100$ (`ModernBERT`), with the sole exception of `NeoBERT`, which fails to converge reliably in either setting. It also stabilises training: the standard deviation of `deberta-v3-base` falls from 0.061 to 0.022.

Decoder-only models behave in the opposite way. `Llama`, `gpt-oss` and `Mistral` each lose between 0.126 and 0.193 macro F1 once retrieved context is added, despite competitive baselines.

Fallacy Classification. The strongest baseline model is `RooseBERT-cont-cased` (0.653 macro F1), followed by `longformer` (0.633) and `RooseBERT-scr-cased` (0.630). Five of the fourteen models fall below 0.35, and baseline variance is markedly higher than in detection: two models exceed a standard deviation of 0.2 across seeds (`roberta` 0.250, `legalbert` 0.237).

The best configuration again uses `Sentence-RooseBERT` with hybrid search, cross-encoder reranking and the *sentence+relations+labels* query, while temporal filtering is not applied. Under it, `ModernBERT` reaches $\mathbf{0.725}$ macro F1, $+0.148$ over its own baseline and $+0.072$ over the best baseline model ($t(4) = 4.62$, $p = 0.010$, $d_z = 2.07$). Retrieval therefore helps on classification as well as on detection, though the margin is narrower.

Part of the gain comes from stabilising weak models: `roberta` improves by $+0.335$ while its standard deviation falls, and `legalbert` gains $+0.069$ with standard deviation down from 0.237 to 0.037. Retrieved context appears to supply discriminative signal that the target sentence alone does not carry for fine-grained types[10].

Decoder-only models degrade sharply. `Llama` loses 0.329 macro F1 and `Mistral` 0.139. Inspection of per-class scores shows this is not a single-class collapse but uniformly low performance across all six fallacy types. The retrieved passages are long, domain-heavy, and unsupervised at inference time. A fine-tuned encoder learns which parts of them matter, a prompted model is simply crowded out.

Error Analysis . We conducted an error analysis comparing the best baseline against the best retrieval model on both tasks (`RooseBERT-scr-cased` vs `ModernBERT` for detection and `RooseBERT-cont-cased` vs `ModernBERT` for classification). Detection and classification benefit from retrieval differently. Temporal filtering helps detection but hurts classification, suggesting that deciding *whether* an argument is fallacious relies on contemporaneous factual context, while classifying *which* fallacy depends on rhetorical form, largely era-independent. Restricting the knowledge base thus removes useful exemplars for classification without providing relevant context.

For the detection, our findings show that `RooseBERT` is biased towards the positive class: 14.98% of the test set consists of non-fallacious arguments predicted as fallacious, twice the rate of `ModernBERT` (7.65%). Retrieval reduces both false positives and false negatives by half, 8.87% and 3.67% respectively, for an overall error rate of 12.54% against the 22.63% of the baseline. The gain for the retrieval is largest on the *NoFallacy* class (recall from 0.7 to 0.82) which also raises precision on the *Fallacy* class from 0.74 to 0.84. This suggests that retrieved context is especially useful for disambiguating rhetorically marked, non-fallacious arguments.

In the classification, both models over-predict *Appeal to Emotion*, which makes up 56.4% of gold labels and accounts for 53% of baseline errors and 55% of errors with retrieval. Both models also confuse *Appeal to Authority* with *Appeal to Emotion* at similar rates, suggesting this is not a knowledge gap: speakers often cite authorities emotionally, so the two classes look alike on the surface, and added context does not separate them. Retrieval help classes defined by form rather than content: *Slogans* recall rises from 0.25 to 0.75, and *Slippery Slope* precision improves from 0.40 to 0.57. This comes at a cost: *Ad Hominem* precision drops from 0.86 to 0.69 and *False Cause* recall from 0.80 to 0.50, so accuracy falls by 1.2 points even as macro F1 gains 3.9 points. Retrieval, then, does not reduce errors uniformly: it redistributes them.

[10] We tested the approach of Goffredo et al. (2022) on the ElecDeb60To20 dataset, though obtaining a lower F1 score than the one reported by the authors. This is likely due to the increased number of unpopulated component/relation features in the new dataset version and affect their architecture.

This pattern reflects the nature of our knowledge base, dominated by legislation, history, and general knowledge. Such sources effectively disambiguate fallacious from non-fallacious arguments but provide fewer useful exemplars for fine-grained type classification, which benefits more from examples.

Across both tasks, `Sentence-RooseBERT` consistently outperforms other retrievers, likely because domain-specific pretraining on parliamentary data aligns better with debate discourse than general-purpose embeddings.

7 Ablation Study

The proposed pipeline combines four components: a structure-informed query, a dense retriever, hybrid fusion with cross-encoder re-ranking, and temporal filtering of the KB. To measure what each contributes we change one component at a time and leave the rest untouched (see Table 3).

Configuration	Detection	Classification
Full pipeline	**80.2**	**71.8**
Retrieval query		
→ `sentence+relations`	78.3	61.0
→ target `sentence` only	73.0	66.4
Retriever		
→ BGE	77.8	57.9
→ SBERT	78.6	60.1
→ BM25 (sparse only)	75.7	60.9
↪ + sentence query	72.7	59.1
Fusion and reranking		
→ dense similarity only	76.3	62.1
Temporal filtering		
→ `off` for detection, `on` for classification	77.5	49.8
Retrieval		
→ no retrieval	75.4	65.3

Table 3: Results of the ablation study (Macro F1) with hyperparameters frozen. Single seed (14).

Each ablation is anchored on the strongest model *without* retrieval: `RooseBERT-scr-cased` for detection, `RooseBERT-cont-cased` for classification, with frozen hyperparameters: batch size of 8, learning rate of 5e-5, trained for 5 epochs.

On detection the full pipeline reaches 80.2 macro-F1 against 75.4 without retrieval ($+4.8$). Replacing the relation query with only target sentence is the most damaging change (-7.2), and pushes the system 2.4 points *below* the no-RAG baseline: retrieving only on the sentence is worse than not retrieving at all. The effect is not a property of the embedding model. BM25 as retriever with a structured query reaches 75.7, the same as with the no-RAG baseline, but BM25 retrieval with the target sentence falls to 72.7. Exchanging one dense retriever for another, by contrast, costs only 1.6–2.4 points, and every dense variant stays above the baseline. Dense retrieval is thus necessary but not sufficient: it is worth 4.4 points over BM25 at a matched query, roughly half of what the query itself is worth, and no choice of embedding model recovers the gain once the query loses its argumentative structure. Reranking sit in the same range (-3.9), marginally above the baseline.

On classification the full pipeline reaches 71.8 against 65.3 ($+6.4$) for the baseline. Swapping the dense retriever is the most expensive change to the retrieval stack (-13.8 for BGE, -11.7 for SBERT): on this task no alternative retriever, sparse or dense, reproduces what Sentence-RooseBERT retrieves. The ordering of the query modes also inverts with respect to detection: `sentence+relations` (61.0) is worse than `sentence-only` (66.4), which is the only configuration to stay above the baseline. We read this as evidence that the gain on classification arises from a specific retriever-query pairing rather than from retrieval quality in general.

8 Conclusion

This paper introduces a novel method for fallacy detection and classification relying on a RAG approach to empower both external knowledge and argumentative structural knowledge. Through the constitution of a 15 GB KB assembled around the entities, topics and time span of the ElecDeb60to20 corpus, we evaluated 14 models across 42 retrieval configurations per task, showing that our method outperforms state-of-the-art competitors.

On **fallacy detection**, retrieval improves every encoder we test, raising the best macro F1 from 0.772 to 0.864 ($p < 10^{-4}$, $d = 8.57$): adding the relations to the query is worth 6.2 macro-F1 points on average. On **fallacy classification** the performance rises from 0.653 to 0.725, a significant improvement ($p < 0.01$, $d = 2.07$) that also collapses the seed variance of the least stable baselines.

The proposed grounded retrieval approach also represents a main step toward interpretable fallacy *explanation*: beside predicting fallacy labels, leveraging retrieved context would allow to explain *why* an argument is fallacious, ensuring transparency in automated political discourse analysis.

Limitations

Computational cost forced a three-stage protocol in which the 42 retrieval configurations were explored using only the three strongest baseline models per task, and the winning configuration was then applied to all 14. The chosen configuration is therefore optimal for those three models and merely reasonable for the rest whose optimum may lie elsewhere in the space.

All experiments use one corpus of English-language U.S. presidential debates from 1960 to 2020, and the KB is constructed specifically around it, drawing on U.S. legislative, census and congressional-biography sources. We applied our methodology to a domain close to ours, propaganda detection, with state of the art results. However, a complete shift in the domain would require to collect a different KB.

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A Additional Experiments

To assess whether our approach generalises beyond ElecDeb60to20, we looked for a dataset annotated with fallacies, or with a closely related phenomenon, whose domain overlaps with our knowledge base.

We selected PROPAGANDA (Martino et al., 2019), a corpus of news articles manually annotated at the fragment level with eighteen propaganda techniques. We consider the sentence-level task, i.e., the binary detection of propagandistic sentences (*propaganda*/*no-propaganda*), for which the authors report a macro F1 of 60.98. PROPAGANDA carries no relation layer, but every sentence is linked to the news article it comes from, which gives us the candidate components needed to recover relations automatically.

We thus trained a relation classifier on the most similar relation-annotated corpus we could find and applied it to PROPAGANDA. That corpus is LIARArg (Wang et al., 2025), 2,832 news claims paired with their justifications, annotated with argument components (claim, premise) and fine-grained relations (support, attack, partial support, partial attack). Following the original work, we merge

the partial labels into *support* and *attack*, yielding a three-way *support/attack/no_relation* task. We trained all the models of Section 5.2 on LIARArg and selected the one with the highest macro F1 averaged over 5 seeds. Results are reported in Table 4.

Models	**Avg. Macro F1**
Llama-3.1-8B-Instruct	**80.5 ± 0.6**
DeBERTa-v3-base	78.6 ± 0.8
gpt-oss-20b	78.5 ± 0.8
Mistral-7B-Instruct-v0.3	76.6 ± 2.9
ModernBERT-base	76.4 ± 1.0
Longformer-base-4096	76.3 ± 1.2
RoBERTa-base	76.0 ± 0.7
RooseBERT-cont-cased	74.7 ± 0.9
LegalBERT	74.5 ± 1.3
RooseBERT-cont-uncased	74.1 ± 1.2
RooseBERT-scr-cased	73.9 ± 0.7
RooseBERT-scr-uncased	72.5 ± 0.8
NeoBERT	57.1 ± 0.7
Qwen3.5-9B-Base	16.3 ± 14.0

Table 4: Argument Component Relation classification performance (mean ± std over 5 seeds) over LIARArg.

`Llama` proved to be the best relation classifier, so we used it to label PROPAGANDA: each propaganda-annotated sentence was paired with every other sentence of its own article, and the predicted *support* and *attack* relations were then used to build the retrieval queries exactly as in our main experiments.

We tested the configuration that performed best on ElecDeb60to20 (dense retriever, hybrid search, re-ranking, and the *sentence+relations+labels* query) and re-tuned only the retriever, running all three dense models on this dataset: here `Sentence-BERT` turned out to be the best choice. The same protocol described in 5.3 was followed.

Table 5 reports the results. Also in this setting our approach improves over the baseline by 10.4 points of macro F1 (24.52 over the score reported by the original authors), confirming that the benefit of relation-aware retrieval is not specific to ElecDeb60to20. The difference between the best baseline model (`Llama` with 75.1 Macro F1) and the best model with retrieval (`longformer` with 85.5 Macro F1) is significant ($p < 0.001$, $d = 5.471$).

Model	Baseline	RAG	Δ
RoBERTa-base	73.5 ± 0.6	85.0 ± 0.8	+11.5
DeBERTa-v3-base	74.5 ± 0.5	42.7 ± 0.0	−31.8
ModernBERT-base	73.3 ± 0.5	81.3 ± 0.8	+8.0
NeoBERT	53.5 ± 14.7	42.7 ± 0.0	−10.8
LegalBERT	73.0 ± 0.4	84.9 ± 0.3	+11.9
Longformer-base-4096	73.7 ± 0.4	**85.5 ± 1.4**	+11.8
RooseBERT-scr-cased	72.7 ± 0.4	83.6 ± 0.5	+10.9
RooseBERT-scr-uncased	74.1 ± 0.4	83.7 ± 0.8	+9.6
RooseBERT-cont-cased	73.4 ± 0.6	84.2 ± 0.3	+10.8
RooseBERT-cont-uncased	73.2 ± 0.8	76.0 ± 18.6	+2.8
Llama-3.1-8B-Instruct	**75.1 ± 0.8**	46.4 ± 1.9	−28.7
Mistral-7B-Instruct-v0.3	74.6 ± 0.8	42.7 ± 0.0	−31.9
Qwen3.5-9B-Base	47.2 ± 7.9	43.4 ± 0.0	−3.8
gpt-oss-20b	74.3 ± 0.6	43.1 ± 0.1	−31.2

Table 5: Propaganda detection performance (macro-F1, mean ± std over 5 seeds) for baseline models and their RAG-augmented counterparts, with the corresponding Δ improvement.

B Implementation Details: Prompts and Hyper-parameters

In this section we report additional details on the implementation. Section B.2 reports each prompt for the LLMs used in our work while Section B.1 reports the hyperparamters for the best models of Table 2.

B.1 Hyper-parameters

Table 6 reports the best hyperparameters of each model for the tasks of fallacy detection and classification and propaganda detection.

B.2 LLMs Prompts

In this section we report the prompt used to train the LLMs in Zero-Shot and Few-Shot setting. In Few-Shot, we appended to the prompt one example per class derived from the training set.

Fallacy Detection - ElecDeb60to20

You are an expert in argumentation theory and fallacy detection. Your task is to determine whether a given sentence from a political debate contains a fallacy. A fallacy is a deceptive argumentative move that appears valid but violates principles of correct reasoning. Answer with exactly one word: "Yes" if the sentence contains a fallacy, or "No" if it does not.

Fallacy Classification - ElecDeb60to20

You are an expert in argumentation theory and fallacy detection. Your task is to identify which fallacy type a given sentence from a political

	Fallacy Detection						Fallacy Classification						Propaganda Detection					
	Baseline			RAG			Baseline			RAG			Baseline			RAG		
Model	BS	LR	Ep.	BS	LR	Ep.	BS	LR	Ep.	BS	LR	Ep.	BS	LR	Ep.	BS	LR	Ep.
RoBERTa-base	16	$2e^{-5}$	5	8	$3e^{-5}$	5	8	$5e^{-5}$	5	8	$3e^{-5}$	5	8	$2e^{-5}$	3	16	$5e^{-5}$	3
DeBERTa-v3-base	8	$2e^{-5}$	5	16	$2e^{-5}$	3	8	$2e^{-5}$	5	16	$5e^{-5}$	5	16	$2e^{-5}$	3	16	$2e^{-5}$	3
ModernBERT-base	8	$3e^{-5}$	5	8	$5e^{-5}$	5	8	$3e^{-5}$	5	8	$5e^{-5}$	5	16	$2e^{-5}$	5	8	$3e^{-5}$	5
NeoBERT	8	$5e^{-5}$	5	16	$2e^{-5}$	5	8	$3e^{-5}$	5	16	$2e^{-5}$	5	8	$3e^{-5}$	3	16	$5e^{-5}$	3
LegalBERT	8	$5e^{-5}$	5	16	$2e^{-5}$	5	8	$5e^{-5}$	5	16	$5e^{-5}$	5	16	$2e^{-5}$	3	16	$3e^{-5}$	3
Longformer-base-4096	16	$5e^{-5}$	5	16	$5e^{-5}$	3	8	$5e^{-5}$	5	8	$3e^{-5}$	5	8	$2e^{-5}$	3	8	$2e^{-5}$	3
RooseBERT-scr-cased	16	$3e^{-5}$	5	8	$5e^{-5}$	5	8	$5e^{-5}$	5	8	$5e^{-5}$	5	16	$5e^{-5}$	3	16	$5e^{-5}$	5
RooseBERT-scr-uncased	8	$3e^{-5}$	5	8	$5e^{-5}$	5	8	$5e^{-5}$	5	16	$2e^{-5}$	5	16	$3e^{-5}$	3	16	$5e^{-5}$	3
RooseBERT-cont-cased	16	$3e^{-5}$	5	8	$5e^{-5}$	5	8	$5e^{-5}$	5	8	$5e^{-5}$	5	8	$3e^{-5}$	3	8	$3e^{-5}$	3
RooseBERT-cont-uncased	8	$5e^{-5}$	5	8	$3e^{-5}$	5	16	$5e^{-5}$	5	8	$5e^{-5}$	5	8	$2e^{-5}$	3	8	$5e^{-5}$	3
Llama-3.1-8B-Instruct	8	$5e^{-4}$	3	8	$1e^{-4}$	1	8	$3e^{-4}$	3	8	$1e^{-4}$	1	8	$1e^{-4}$	2	8	$3e^{-4}$	3
Mistral-7B-Instruct-v0.3	*zero-shot*			8	$1e^{-4}$	1	8	$3e^{-4}$	2	*zero-shot*			8	$3e^{-4}$	2	*few-shot*		
Qwen3.5-9B-Base	8	$5e^{-4}$	3	8	$5e^{-4}$	2	*few-shot*			8	$1e^{-4}$	1	8	$5e^{-4}$	2	*zero-shot*		
gpt-oss-20b	8	$5e^{-4}$	3	8	$5e^{-4}$	3	8	$5e^{-4}$	1	8	$1e^{-4}$	1	8	$3e^{-4}$	3	8	$5e^{-4}$	1

Table 6: Best hyperparameters for fallacy detection and classification (ElecDeb60to20), and for propaganda detection (PROPAGANDA), Baseline vs. RAG settings (5 seeds). BS = batch size, LR = learning rate, Ep. = num. of epochs.

debate contains. A fallacy is a deceptive argumentative move that appears valid but violates principles of correct reasoning. Consider only these six fallacy types:
- Ad Hominem: an excessive attack on the opponent's character (e.g. name-calling and labelling, tu quoque / "you did it first", or bias) instead of addressing their argument.
- Appeal to Authority: citing an authority or a group's agreement without providing relevant evidence, including appeals to non-experts or to mere popular acceptance.
- Appeal to Emotion: loading the argument with emotional language (appeal to pity, appeal to fear, loaded language, or flag waving) to exploit the audience's feelings.
- False Cause: mistaking a correlation between two events for a causal relationship.
- Slippery Slope: claiming that an act will lead to an unlikely, exaggerated outcome while omitting the intermediate steps. - Slogan: a brief and striking phrase used to provoke excitement, often reinforced by repetition.
Every sentence contains exactly one of these six fallacy types — there is always a fallacy present, so "None" is never a valid answer. Answer with exactly one fallacy type, using exactly one of those six names, and nothing else.

Propaganda Detection - Propaganda

You are an expert in propaganda detection and persuasion analysis. Your task is to determine whether a given sentence from a news article contains a propaganda technique. A propaganda technique is a persuasive device (such as loaded language, name calling, exaggeration, appeal to fear, or flag waving) used to influence an audience beyond sound reasoning. Answer with exactly one word: "Yes" if the sentence contains a propaganda technique, or "No" if it does not.

Relation Classification - LIARArg

You are an expert in argumentation analysis. You are given two sentences drawn from a fact-checking argumentation corpus: a claim (Sentence 1) and a candidate piece of evidence (Sentence 2). Your task is to identify the argumentative relation that Sentence 2 holds toward Sentence 1. Consider only these three relations:
- Support: Sentence 2 provides evidence or reasoning in favour of Sentence 1.
- Attack: Sentence 2 provides evidence or reasoning against Sentence 1.
- No Relation: Sentence 2 is not argumentatively related to Sentence 1.
Answer with exactly one of those three relation names, and nothing else.